from IBM and from various universities. In the test, Google scored the highest amongst all competing software in Arabic-to-English and Chinese-to-English translation tests; these were conducted by the NIST (National Institute of Science and Technology). Each test comprised the task of translating a hundred articles from Agence France Presse and the Chinese Xinhua News Agency from December 2004 to January 2005.

How?

The answer lies in statistical analysis. It now seems increasingly likely that rather than have a system try to understand a piece of text, or formulate abstract representations taking context and other things into account, the most promising method of translation involves looking at ready translations. How this works is, the system would look at existing translations, and be trained on those. For example, if the German "reich" is often seen as being translated as "rich" *in the context of money*, the system would pick it up. And "Reich" gets translated as "realm" or "rule" in the appropriate context—as in "Die dritte Reich" (remember "The Rise and Fall of the Third Reich"?)—and the distinction would be made.

This means that the more the number of documents in the source and target languages available, the better the system would get at translating. And, of course, Google has a very large store of documents in various languages.

Translation is a hard AI endeavour, and it's unlikely we'll see a good automated translator soon

Humans To The Rescue

Chess is an example of an AI endeavour that has succeeded remarkably well. Most others don't. Translation, like we said, is a hard AI endeavour, and it's unlikely we'll see a good automated translator soon. But there are three things to consider here. First off, to aid a machine in its job, the source text may have to be pre-edited, or "normalised," to make it suitable for translation, as in the following:

I know I'm wrong.

I know, I'm wrong.

The two in English are similar, but the translations into German by free online translators result in two very different things! Someone who (a) knows the idiosyncrasies of the translation system, and (b) has a good knowledge of both languages, could "format" text for the best results. Pre-editing is increasingly being seen as an important step. One can come up with several examples where pre-editing would be required—slang would need to be toned down, casual utterances would need to be formalised, missing words such as "that" (as in "I know *that* I'm wrong") would be inserted, and so on.

Experts In Their Field

The second thing we need to look at in the context of not ever getting good machine translators is that of domain-specific translation. "Domain," here, refers to fields such as weather forecasting, physics, and so on. Why we mention weather in particular is that in Canada, the Météo system has been used to translate weather reports from English to French. It has worked well for decades now, and is still being used without anyone even noticing or complaining! How come? Simply because the system has been tailor-made for the field, and because the terminology rarely changes. There's no slang; the words always have a fixed meaning—for example, a "cold front" is always in the weather sense of the phrase.

Thus, instead of aiming at general-purpose translation systems, we could set our sights lower and think of developing several domain-specific systems. Thankfully, a lot of the material on the Internet that needs to be translated is domain-specific.

Helping Hands

As a third point, we gather that in the face of the problem being hard for AI, the future might well see the machine translator working as an aid in the translation process. In the case of the latter, we can envisage someone with less-than-perfect knowledge of the target language being able to translate reasonably well with the help of a machine: the machine would do a lot of the dictionary searching and basic grammar construction, while the human would do the dirty work of supplying context, and re-consulting the machine in the case of an ambiguity. However, it all boils down to money: critics of this approach say it would be just as expensive as human translation, but proponents say it would be less expensive on account of the number of man-hours spent.

One other approach being seen as the future of translation is that human translators would proofread machine translations. This, as in the former case, has the advantage of reducing the man-hours spent in the translation process. It has the added advantage of the final output having been read and verified by a human.

Is This The Future?

We've painted a somewhat anaemic picture so far—that of MT not being up to speed, and fully automated translation not being possible in the near future. But here's an excerpt from post-gazette.com, dated 28 October 2005:

"Stan Jou's lips were moving, but no sound was coming out. Mr Jou, a graduate student in language technologies at Carnegie Mellon University, was simply mouthing words in his native Mandarin Chinese. But 11 electrodes attached to his face and neck detected his

Some Online Translation Sites

Site	Comments
www.systransoft.com	Reasonably good; one of the oldest existing tools
www.google.com/language_tools	Probably the best one out there
http://babelfish.altavista.com/tr	Quirky. Sometimes OK, sometimes pathetic
http://trans.voila.fr/voila	Site is in French; works better for French translations
www.linguatec.net/online/ptwebtext/index_en.shtml	Has an excellent feature: subject area selection for more accurate translation. Works better for German translations
http://translation2.paralink.com/	Poor, even at German translation, which is the site's speciality
www.reverso.net/text_translation.asp	Very bad, even though the site seems to have won several awards